

**Total No of Questions: [8]**

**SEAT NO. :**

**[Total No. of Pages : 2 ]**

***S.E. 2012 (Computer Engineering)***  
***Computer Organization***  
***(Semester -II)***

***Time: 2Hours***

**Max. Marks : 50**

**Instructions to the candidates:**

- 1) **Answers Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.**
- 2) **Neat diagrams must be drawn wherever necessary.**
- 3) **Figures to the right side indicate full marks.**
- 4) **Use of Calculator is allowed.**
- 5) **Assume Suitable data if necessary**

- Q1) a) Using Booth's algorithm multiply the following [6]  
Multiplicand = 7, Multiplier = 3
- b) Explain IEEE 488 format for single precision and double precision floating point numbers with example. [6]

OR

- |     |    |                                                                   |     |
|-----|----|-------------------------------------------------------------------|-----|
| Q2) | a) | Explain the various speeds up techniques of processor             | [4] |
|     | b) | Explain following addressing modes of 8086 with suitable examples | [8] |
|     |    | a. Index Addressing                                               |     |
|     |    | b. Register Indirect                                              |     |
|     |    | c. Auto Increment                                                 |     |
|     |    | d. Relative Addressing                                            |     |

- |     |    |                                                                         |     |
|-----|----|-------------------------------------------------------------------------|-----|
| Q3) | a) | Draw flowchart for Non restoring Division algorithm                     | [6] |
|     | b) | Write control sequence for execution of the instruction<br>Add (R3), R1 | [6] |

OR

- Q4) a) Explain the Key components of the front end of the Intel Nehalem architecture. [6]  
b) What are the different design methods for Hardwired control units? Explain any one. [6]

- |     |    |                                                        |     |
|-----|----|--------------------------------------------------------|-----|
| Q5) | a) | Explain cache mapping techniques with example          | [6] |
|     | b) | Explain Intel Nehalem memory organization with diagram | [7] |

OR

- |     |                               |      |
|-----|-------------------------------|------|
| Q6) | Write short Notes (Any Three) | [13] |
|     | DDR3 Memory                   |      |
|     | NUMA                          |      |

USB Packet Format

PCI Bus

Q7) a) Draw and explain block diagram of itanium processor [6]

b) Write short note on [7]

1. i7 Mobile Version

2. Instruction format of IA-64 architecture

OR

Q8) a) Explain the architecture of CBE processor with the help of block diagram [6]

b) Write short note on [7]

1. Sun UltraSparc T1

2. NVIDIA GPU